Concrete Strength Report

This report describes the Concrete Compressive Strength dataset, the preparation applied to it and the baseline used to evaluate later models. The dataset contained 1,030 laboratory tests with no missing values. The 25 exact duplicate rows were removed, which left 1,005 observations. These were divided into a training set of 804 rows and a test set of 201 rows. A mean-prediction baseline was then fitted. On the test set it achieved an **RMSE of 17.3 MPa, an MAE of 14.0 MPa and an R^2 of 0.00**. These values were adopted as the minimum standard for all subsequent models.

1. The data

The analysis used the UCI Concrete Compressive Strength dataset (Yeh, 1998). Each observation records the composition of one concrete mix, the curing age at testing and the measured compressive strength. The dataset has eight numeric predictors and one continuous target, so the task was treated as regression. The variables are summarised below.

Variable	Unit	Mean	Min to max	Note
Cement	kg/m³	281	102 to 540	Main binder
Blast furnace slag	kg/m³	74	0 to 359	Zero in 46% of mixes
Fly ash	kg/m³	54	0 to 200	Zero in 55% of mixes
Water	kg/m³	182	122 to 247	
Superplasticizer	kg/m³	6.2	0 to 32	Zero in 37% of mixes
Coarse aggregate	kg/m³	973	801 to 1,145	
Fine aggregate	kg/m³	774	594 to 993	
Age	days	46	1 to 365	14 discrete values, 41% at 28 days
Strength (target)	MPa	35.8	2.3 to 82.6	Standard deviation 16.7

The 1,030 observations were found to represent only 427 distinct mixes, since most mixes were tested at more than one age.

2. Data quality findings

Check	Result
Missing values	No missing values were found in any column.
Exact duplicates	25 rows (2.4%) were identical to another row on all nine columns.
Repeated tests	Nine mix and age combinations had been tested more than once with different results. The median difference between

Check	Result
	repeats was 5.8 MPa, which is consistent with normal laboratory variation.
Zero values	Slag, fly ash and superplasticizer were zero in 46%, 55% and 37% of mixes. These zeros indicate that the ingredient was not used, not that a value was missing.
Outliers (1.5 × IQR rule)	Outliers were flagged in age (59, all tests at 180 days or more), superplasticizer (10), water (9), fine aggregate (5), strength (4) and slag (2). Each flagged value was within the normal range of concrete practice.
Physical consistency	The total mass of each mix ranged from 2,195 to 2,551 kg/m³, which agrees with the density of real concrete. No data entry errors were detected.
Skewness	Age was strongly right-skewed (skewness 3.3). The other variables showed only mild skew (below 1).
Correlation with strength	Strength was positively correlated with cement (r = 0.50), superplasticizer (0.37) and age (0.33), and negatively correlated with water (−0.29). Two derived measures were more strongly related to strength than the raw columns: log(age) (0.55) and the water/cement ratio (−0.50).
Multicollinearity	The strongest correlation between two predictors was between water and superplasticizer (−0.66). No pair was judged high enough to cause concern.

Limitations. The data originates from a single laboratory source. Few tests were conducted beyond 90 days, and the dataset does not record cement type, temperature or curing method.

3. Data preparation

Step	Action taken	Justification
1. Duplicates	The 25 exact duplicates were removed, leaving 1,005 rows.	Duplicates could appear in both the training and test sets, allowing a model to score well by memorisation and inflating its results.
2. Repeated tests	The nine sets of repeated tests were retained.	They are independent measurements rather than copies, and they reflect the level of noise that no model can remove.
3. Zero values	Zeros were kept as recorded.	They represent genuine mix design choices. Imputing them would have introduced ingredients that were never used.

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4. Outliers	All outliers were retained.	Every flagged value was physically valid. Removing the long-age tests would have eliminated the only evidence on long-term strength development.
5. Feature engineering (linear models only)	log(age) and the water/cement ratio were added, and raw age was removed.	log(age) reflects the curing curve, in which strength rises quickly up to 28 days and then levels off. The water/cement ratio follows Abrams' law, under which strength decreases as the ratio increases. Both features were more strongly correlated with strength than the raw columns. Tree-based models received the raw inputs because they can capture these patterns directly.
6. Scaling	Predictors were standardised within a pipeline for the linear models.	Input values ranged from about 6 to 1,000 kg/m³. Fitting the scaler inside the pipeline ensured it used training folds only, so no test information leaked into training.
7. Train/test split	20% of the data was held out as a test set (random seed 42). 5-fold cross-validation was applied to the remaining 80%.	The test set was used only once, for final evaluation. Cross-validation provided a fair comparison between models during development.

Check for leakage between mixes. Because most mixes were tested at several ages, a random split placed the same mix in both the training and test sets. To measure the effect, a second split was made that grouped rows by mix, so that no mix appeared in both sets (GroupShuffleSplit and GroupKFold). The results are reported in Section 4.

4. Mean-prediction baseline

A mean-prediction baseline was established using scikit-learn's `DummyRegressor(strategy="mean")`. The model ignored all predictors and predicted the training-set mean strength of 35.1 MPa for every mix. This measured the error that results from guessing the average alone.

Evaluation	RMSE (MPa)	MAE (MPa)	R^2
5-fold cross-validation, training data	16.0 (sd 0.6)	13.0	−0.01
Hold-out test set, random split (201 rows)	17.3	14.0	0.00
Test set over 10 random splits	16.4 (sd 0.4), range 15.4 to 17.1		
Hold-out test set, split by mix (192 rows)	15.6	12.6	−0.01
5-fold cross-validation, split by mix	16.3 (sd 1.3)		−0.01

Interpretation. The baseline RMSE was close to the standard deviation of strength (16.7 MPa), and the R^2 of zero confirmed that it explained none of the variation in strength. A median baseline was also tested and produced almost the same result (test RMSE 17.4 MPa). The grouped split gave similar baseline scores, so mix leakage did not affect the baseline. A test RMSE of 17.3 MPa and an R^2 of 0 were therefore adopted as the reference against which all subsequent models were evaluated.